Ewoud

Electronics & PCB design

Massachusetts · github.com/EwoudVV · ewoudvv.github.io

10th-grade student focused on electronics, PCB design, and hardware integration. Seeking a summer 2027 electronics engineering internship, with an interest in MIT Lincoln Laboratory's Group 76 software and electronics work.

EDUCATION

Lincoln-Sudbury Regional High School
Expected graduation: 2029 · Massachusetts

ENGINEERING PROJECTS

Ducktop2 | Custom 16-inch laptop

Main-board routing in progress

- Designing an eight-layer center motherboard and I/O boards around the LattePanda Mu Ultra, with PCIe NVMe, USB, HDMI, Ethernet, and a separate Wi-Fi/Bluetooth socket.
- Working on differential-pair routing, power distribution, battery integration, and mechanical clearances. The complete laptop has not reached powered integration.

STM32 development board

Design complete; awaiting fabrication

- Designed a 24 × 44 mm, four-layer STM32F411 board with USB-C, SPI microSD, 22 GPIO pins, and SWD access.
- Prepared schematics, layout, Gerbers, BOM, placement files, and a pinout. Starter firmware builds; assembled-hardware testing is pending.

Battery management & charging | Ducktop2 subsystem

BMS routed; bench tests pending

- Designed a four-layer, three-cell BMS using a BQ76922 monitor, high-side protection MOSFETs, a four-terminal current shunt, and independent current guards.
- Integrating an ISL9241 charger on the center board. Current capacity, fault response, and charging behavior still need hardware qualification.

BLDC motor controller

PCB routed; not built or powered

- Designed a four-layer STM32G474 controller with a DRV8353S gate driver, six-MOSFET inverter, phase-current sensing, and protection circuitry.
- Prepared fabrication files and bring-up firmware. Hall-based control is planned for a 26 V Groschopp gearmotor; commutation and FOC are not implemented.

TECHNICAL SKILLS

KiCad · Schematic design · Multilayer PCB layout · Differential-pair routing · Component selection · Power electronics · Microcontroller hardware · Git/GitHub · Linux · Basic Python and C#

COMMUNITY

Hack Club member · Half Life hardware reviewer